REPORT

Asia oil price comparison

1. Tax Pass-Through Elasticity

How much of global fuel price change is passed to retail prices across countries?

Methodology: price_effect or pass through is calculated using change in retail price (per liter)/ change in crude price(per liter).

Price_effect	Meaning
≈ 1	Full transmission
< 1	Partial Absorption
> 1	Amplification

country	year_date	price_effect
China	2016	2.16
China	2017	2.28
China	2018	2.00
China	2019	1.83
China	2020	2.25
China	2021	2.28
China	2022	2.17
China	2023	2.11
China	2024	2.29
China	2025	2.33
China	2026	2.30
India	2016	2.07
India	2017	2.03
India	2018	1.97

2. Subsidy Cushion Effect

Do higher fuel subsidies reduce the transmission and volatility of global oil price changes in domestic fuel markets?

Methodology: Retail-to-crude ratio is used as a proxy for pricing burden.

country	retail/global(usd)	gasoline_tax_pct	subsidy_pct_gdp
China	2.23	26	0.00
India	2.01	52	2.10
Indonesia	1.33	5	7.80
Japan	2.79	42	4.50
Malaysia	0.89	0	6.50
Pakistan	1.17	17	2.00
Thailand	2.02	27	2.50

The analysis reveals that gasoline taxes play a dominant role in determining retail fuel prices across countries, while subsidies have a more limited and inconsistent effect. Countries with high tax rates, such as India and Japan, exhibit significantly higher retail-to-crude price ratios despite substantial subsidy expenditure. In contrast, countries like Malaysia, which combine low taxes with high subsidies, maintain the lowest fuel prices. This suggests that taxation policy outweighs subsidy policy in shaping final fuel prices, and that subsidies alone are insufficient to offset the price-increasing effects of high taxes.

3. Price Convergence / Divergence

Are fuel prices across the 12 countries converging or diverging over time?

Methodology: Dispersion is measured as the standard deviation of country-level average prices relative to the global mean.

year_date	divergence(stddev)
2015	0.04
2016	0.03
2017	0.04
2018	0.07
2019	0.06
2020	0.03
2021	0.07
2022	0.14
2023	0.1
2024	0.09
2025	0.06
2026	0.1

this table shows the difference in yearly average fuel price of each country to the global average of every year using standard deviation yearly. The standard deviation of fuel prices across countries indicates that price dispersion was relatively low and stable between 2015 and 2020, suggesting mild convergence. However, a significant increase in dispersion is observed in 2021–2022, indicating strong divergence in fuel pricing across countries. Although dispersion decreases after 2022, it remains above early-period levels, suggesting only partial re-convergence and increased volatility in recent years.

4. Volatility Decomposition

Is price volatility driven more by global price movement or domestic policy (tax/subsidy)?

Methodology: Relative volatility is measured as standard deviation normalized by mean price.

China	2025	0.11	0.11	-0.49
China	2026	0.27	0.26	-0.63
India	2015	0.15	0.15	-0.34
India	2016	0.16	0.15	-0.28
India	2017	0.1	0.09	-0.35
India	2018	0.11	0.09	-0.45
India	2019	0.07	0.06	-0.41
India	2020	0.21	0.23	-0.28
India	2021	0.12	0.11	-0.44
India	2022	0.11	0.12	-0.63
India	2023	0.06	0.07	-0.52
India	2024	0.06	0.06	-0.51
India	2025	0.1	0.11	-0.41
India	2026	0.24	0.26	-0.51
Indonesia	2015	0.13	0.15	-0.13

dev_brent ≈ dev_gasoline	global price movement dominates
dev_gasoline > dev_brent	domestic factors amplify volatility
dev_gasoline >> dev_brent consistently	strong domestic policy/supply effects and vice versa

5. Petrol vs Diesel Policy Bias

Which countries systematically favor diesel over petrol (or vice versa) through taxation?

Methodology: Policy bias is measured as the average difference between gasoline and diesel tax rates.

country	avg_tax_difference	policy_bias
South Korea	14.000	Favors Diesel (higher petrol tax)
Thailand	12.000	Favors Diesel (higher petrol tax)
Singapore	7.000	Favors Diesel (higher petrol tax)
India	5.000	Favors Diesel (higher petrol tax)
Bangladesh	4.000	Favors Diesel (higher petrol tax)
Japan	4.000	Favors Diesel (higher petrol tax)
Vietnam	4.000	Favors Diesel (higher petrol tax)
China	2.000	Favors Diesel (higher petrol tax)
Philippines	2.000	Favors Diesel (higher petrol tax)

6. Outlier Detection (Policy Shock)

Identify dates where domestic prices deviate significantly from expected values based on global trends.

Methodology: Countries are compared against their tax-category average price.

tax_burden_category	country	gasoline_usd_per_liter	category_avg	diff_gasoline
High	Japan	1.15	1.21	-0.06
High	India	1.11	1.21	-0.10
Moderate	China	1.26	1.44	-0.18
Moderate	Thailand	1.24	1.44	-0.20
Moderate	Vietnam	1.09	1.44	-0.35
Moderate	Bangladesh	1.07	1.44	-0.37

7. Rolling Volatility (Time-Series)

How does rolling 3-month or 6-month volatility differ across countries?

Methodology: Rolling volatility is computed using a moving window of 3 and 6 months.

country	year(dates)	month(dates)	rolling_dev_3months	rolling_dev_6months
China	2026	2	NULL	NULL
China	2026	3	0.22	NULL
China	2026	4	0.28	NULL
India	2015	1	NULL	NULL
India	2015	2	NULL	NULL
India	2015	3	0.06	NULL
India	2015	4	0.04	NULL
India	2015	5	0.05	NULL
India	2015	6	0.02	0.07
India	2015	7	0.04	0.04
India	2015	8	0.06	0.07
India	2015	9	0.04	0.08
India	2015	10	0.01	0.08
India	2015	11	0.03	0.07
India	2015	12	0.06	0.07
India	2016	1	NULL	NULL
India	2016	2	NULL	NULL

8. Shock Absorption Index

Which countries absorb global shocks vs transmit them fully in 2026?

Methodology: Shock transmission is measured as the difference between monthly changes in retail and crude prices.

country	dates	change_in_retail_price	change_in_crude_price	diff_in_changes
China	2026-04-01	0.59	0.26	0.33
Thailand	2026-04-01	0.63	0.32	0.31
Japan	2026-04-01	0.27	0.10	0.17
Indonesia	2026-04-01	0.40	0.29	0.11
India	2026-04-01	0.17	0.10	0.07
Pakistan	2026-04-01	0.32	0.30	0.02
Malaysia	2026-04-01	0.08	0.10	-0.02